

Exact qutrit counterexamples to Conjecture 1 of Das and Mori

Dehao Lin

ORCID: 0009-0001-4551-8490

School of Physics, Sun Yat-sen University, Guangzhou, China

Das and Mori's Conjecture 1 proposes the two-sided hierarchy (their Eq. (4)) in the conventions of the full manuscript:

$$\mathcal{C}_S(\Psi_\rho^{U*}(t)) \leq \mathcal{C}_K(\rho(t)) \leq \mathcal{C}_K(\Psi_\rho^I(t)).$$

The two sides are violated by two separate strictly full-rank qutrit witnesses, given below. No single displayed state is claimed to violate both sides. These are the frozen results of Propositions 3.1 and 3.2, not new results.

Write $S_{U*} = \mathcal{C}_S(\Psi_\rho^{U*}(t))$, $C_M = \mathcal{C}_K(\rho(t))$, and $K_I = \mathcal{C}_K(\Psi_\rho^I(t))$. Here $\rho(t) = e^{-itH}\rho e^{itH}$ and $P = \text{Tr } \rho^2$. Operator complexity uses the Hilbert–Schmidt inner product and normalized mixed seed $\rho/\sqrt{P}$. For $s = \text{vec}_r \sqrt{\rho}$, the purification generators are $G_I = H \otimes I$ and $G_{U*} = H \otimes I - I \otimes \bar{H}$. The rightmost quantity K_I is the operator complexity of the rank-one density matrix $ss^\dagger$ under $[G_I, \cdot]$, not its vector spread complexity.

A. Lower-side counterexample

$$H_L = \text{diag}(2, 0, 1), \quad \rho_L = \frac{1}{42} \begin{pmatrix} 32 & 0 & 0 \\ 0 & 5 & 3 \\ 0 & 3 & 5 \end{pmatrix}.$$

Its spectrum is $16/21, 4/21, 1/21$, all positive. The square root and purity are

$$\sqrt{\rho_L} = \frac{1}{2\sqrt{21}} \begin{pmatrix} 8 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \quad P_L = \frac{13}{21}.$$

Only levels two and three carry coherence, with energy gap one. In the energy basis, the mixed and spread spectral weights are respectively $|\rho_{ab}|^2/P$ and $|(\sqrt{\rho})_{ab}|^2$ at $E_a - E_b$. Both measures therefore have the three-atom form $\nu_\mu = (1 - \mu)\delta_0 + (\mu/2)(\delta_{-1} + \delta_1)$, with

$$\mu_S = \frac{1}{42}, \quad \mu_K = \frac{3}{182}, \quad S_{U*}(t) = F(\mu_S; t), \quad C_M(t) = F(\mu_K; t).$$

Lemma 2.1 gives the exact three-atom formula, for $0 < \mu < 1$,

$$F(\mu; t) = \mu \sin^2 t + 8\mu(1 - \mu) \sin^4(t/2).$$

For a direct check, the three Krylov amplitudes are $(1 - \mu + \mu \cos t, -i\sqrt{\mu} \sin t, \sqrt{\mu(1 - \mu)}(\cos t - 1))$; their squared moduli, weighted by indices 0, 1, 2, give F . At $t = \pi$ this yields the exact violation

$$S_{U*}(\pi) = \frac{82}{441}, \quad C_M(\pi) = \frac{1074}{8281}, \quad \frac{82}{441} - \frac{1074}{8281} = \frac{4192}{74529} > 0.$$

Thus $S_{U*}(\pi) > C_M(\pi)$, violating the lower side of the proposed hierarchy.

B. Upper-side counterexample

$$\rho_R = \frac{1}{26} \text{diag}(16, 1, 9), \quad H_R = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This state is strictly full rank, with purity $P_R = 1/2$ and square root $\text{diag}(4, 1, 3)/\sqrt{26}$. The two operator problems have the following spectral measures; each weight corresponds to the support entry in the same position:

	ordered support	ordered weights
C_M	$(-2, 0, 2)$	$(225/1352, 451/676, 225/1352)$
K_I	$(-2, -1, 0, 1, 2)$	$(289/2704, 153/676, 451/1352, 153/676, 289/2704)$

For K_I , the purified vector has energy weights $17/52, 9/26, 17/52$ on $-1, 0, 1$; their difference convolution gives the second row. All weights are positive, so the Krylov chains have exact lengths three and five.

To check the evolution from either row, orthonormalize $1, x, x^2, \dots$ against its spectral measure ν to obtain real polynomials p_n . The frozen spectral prescription and complexity definition are

$$\varphi_n(t) = \int p_n(x) e^{-itx} d\nu(x), \quad C(t) = \sum_{n=0}^{m-1} n |\varphi_n(t)|^2.$$

For additional checks on this finite calculation, the squared nonzero Lanczos coefficients are $(225/169, 451/169)$ for C_M and $(17/13, 43/26, 1377/1118, 451/559)$ for K_I ; all diagonal coefficients vanish. At $t = \pi/3$, Appendix A gives the complete probabilities in increasing Krylov index:

$$p^M = \left(\frac{458329}{1827904}, \frac{675}{2704}, \frac{913275}{1827904} \right),$$

$$p^{K,I} = \left(\frac{1500625}{7311616}, \frac{62475}{140608}, \frac{45778977}{157199744}, \frac{7803}{140608}, \frac{1173051}{314399488} \right).$$

Each row sums to one. Its index-weighted sum gives, exactly,

$$C_M(\pi/3) = \frac{1141425}{913952}, \quad K_I(\pi/3) = \frac{2967537}{2456246}, \quad C_M(\pi/3) - K_I(\pi/3) = \frac{1600683}{39299936} > 0.$$

Thus $C_M(\pi/3) > K_I(\pi/3)$, violating the upper side. As an independent sanity check, the frozen short-time coefficients are

$$C_M(t) = \frac{225}{169} t^2 + O(t^4), \quad K_I(t) = \frac{221}{169} t^2 + O(t^4).$$

The full manuscript also establishes the violation throughout $0 < t < 1/2704$.

Minimality and scope

Both displayed states are strictly full rank; they are separate witnesses for separate sides. Theorem 3.3 states: For every qubit density matrix, every time-independent Hermitian Hamiltonian, and every real time, $S_{U^*}(t) \leq C_M(t) \leq K_I(t)$. Consequently, physical dimension three is minimal for failure of either side of the source hierarchy.

This note is a communication derivative of the full manuscript *Exact limits of Krylov-complexity comparisons under purification*. The complete proof context, AI-contribution disclosure, exact certificates, audits, and reproducibility materials are provided in the accompanying repository.